

Balance, Threshold Calibration, and Adversarial Neighbors on the Sphere

with Applications to Random Deep Networks

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Abstract

We isolate a single geometric parameter governing the existence of nearby adversarial points on a high-dimensional sphere: the smaller class mass. If a measurable classifier $F : \mathbb{S}^{d-1}(R) \rightarrow \{\pm 1\}$ is p -balanced, then for a $1 - \delta$ fraction of points x there exists y on the same sphere with $F(y) \neq F(x)$ and

$$\|x - y\| \lesssim \frac{R}{\sqrt{d}} \left(\sqrt{\log(1/p)} + \sqrt{\log(1/\delta)} \right).$$

This is a direct reformulation of spherical concentration and isoperimetry.

We then study how balance arises from network structure. Bias-free odd architectures (including odd fully connected nets, CNNs, residual networks, and a bias-free odd ViT model) are exactly balanced once their zero level set is null. For general scores, we separate three notions that are often conflated: exact balance produced by a genuine scalar threshold shift, exact balance produced by measurable splitting of a level set, and mere nontriviality of the deployed classifier. A sharp criterion shows that a scalar threshold shift yields exact balance if and only if the pushforward score distribution hits the $1/2$ quantile without a jump; atomless score distributions suffice, while ReLU flat regions can obstruct exact balance. Finally, for random Gaussian readouts over a fixed feature map, the positive-class mass has mean $1/2$ and variance controlled by an average feature-correlation parameter, yielding near-balance and therefore adversarial neighbors for random heads.

The paper is intended as a balance-based synthesis of known adversarial-existence phenomena rather than a new proof of the optimal $R/\sqrt{d}$ scaling.

1 Introduction

Adversarial examples in high dimensions are by now known to arise from several different mechanisms. One line of work is geometric: concentration of measure on spheres, cubes, and related spaces forces large sets to have thick boundaries, which in turn produces nearby label flips for many points [7, 13]. Another line studies random neural networks directly. For fully connected random ReLU networks, adversarial examples and even gradient-based findability were established by Daniely and Shacham [5], Bubeck et al. [2], Bartlett et al. [1], and Montanari and Wu [12]. For random CNNs, Daniely [3] derived adversarial examples from isoperimetry on $SO(d)$, and Daniely and Mehalel

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[4] recently proved essentially optimal ℓ_2 perturbation size $\Theta(\|x\|/\sqrt{d})$ together with a one-step gradient construction. The Gaussian-process work of De Palma et al. [6] also identifies the same $1/\sqrt{d}$ boundary-distance scaling.

The present paper is a shorter and more conservative rewrite of an earlier draft. Its main point is conceptual:

once a classifier uses both labels on a sphere in non-negligible proportion, spherical concentration already forces nearby adversarial points.

The network enters only through the balance parameter

$$p := \min\{\mu(F = +1), \mu(F = -1)\}.$$

Everything geometric follows from p and the dimension.

We emphasize two corrections relative to the earlier draft. First, exact balance obtained by *splitting a level set* is not the same thing as exact balance obtained by a *scalar threshold shift* of the logit. The former is always possible at a median on a nonatomic space; the latter can fail when the score distribution has an atom at the median, which happens naturally for ReLU networks with flat regions. Second, a previously stated measure-theoretic “rich feature” condition for random Gaussian heads was too weak; here we replace it with a clean deterministic hypothesis on the feature image, namely $0 \in \text{int conv } a(\mathbb{S}^{d-1}(R))$. Under continuity, that hypothesis guarantees that every nonzero readout uses both labels on positive measure.

The $R/\sqrt{d}$ scaling is not new. What is new here is the organization:

- a sharp balance-based theorem on the sphere with explicit dependence on p and δ ;
- a precise criterion for when exact balance can be produced by a genuine scalar threshold shift;
- a clean separation between oddness, threshold calibration, and random-readout arguments;
- a modest transfer principle from the uniform spherical measure to any bounded-density measure on the same sphere.

This makes the paper more accurate, shorter, and easier to position relative to the existing literature.

2 Geometry on the sphere

Fix $d \geq 3$ and $R > 0$. We work on the sphere

$$\mathbb{S}^{d-1}(R) := \{x \in \mathbb{R}^d : \|x\| = R\}$$

with its normalized rotation-invariant measure μ . For a measurable set $A \subseteq \mathbb{S}^{d-1}(R)$ and $\varepsilon > 0$, write

$$A_\varepsilon := \{x \in \mathbb{S}^{d-1}(R) : \text{dist}(x, A) \leq \varepsilon\}, \quad \text{dist}(x, A) := \inf_{y \in A} \|x - y\|.$$

2.1 A quantitative thickening lemma

We use the following one-sided form of Lévy concentration around a median, a standard consequence of sharp spherical isoperimetry; see, for example, Milman [11, Theorem 1.1] or Ledoux [9, Chapter 3].

Lemma 1 (One-sided concentration around a median). *Let $h : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ be 1-Lipschitz with respect to the ambient Euclidean metric, and let m_h be any median of h under μ . Then for every $t > 0$,*

$$\mu(h \geq m_h + t) \leq C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right), \quad C_{\text{Lev}} := \sqrt{\frac{\pi}{8}}, \quad (2.1)$$

and similarly

$$\mu(h \leq m_h - t) \leq C_{\text{Lev}} \exp\left(-\frac{(d-1)t^2}{2R^2}\right). \quad (2.2)$$

Lemma 2 (Thickening on the sphere). *Let $A \subseteq \mathbb{S}^{d-1}(R)$ be measurable with $\mu(A) = \alpha \in (0, 1)$. Define*

$$r(\alpha) := \begin{cases} 0, & \alpha \geq 1/2, \\ \frac{R}{\sqrt{d-1}} \sqrt{2 \log \frac{1}{\alpha}}, & \alpha < 1/2. \end{cases} \quad (2.3)$$

Then for every $\varepsilon > 0$,

$$\mu(A_\varepsilon) \geq 1 - C_{\text{Lev}} \exp\left(-\frac{d-1}{2R^2} (\varepsilon - r(\alpha))^2_+\right), \quad (2.4)$$

where $(u)_+ := \max\{u, 0\}$.

Proof. Let $h(x) := \text{dist}(x, A)$. Then h is 1-Lipschitz and vanishes on A . Let m be a median of h . If $\alpha \geq 1/2$, then $\mu(h = 0) = \alpha \geq 1/2$, so $m = 0$.

Assume now that $\alpha < 1/2$. Since $\mu(h \leq 0) = \alpha$, Lemma 1 gives

$$\alpha = \mu(h \leq m - m) \leq C_{\text{Lev}} \exp\left(-\frac{(d-1)m^2}{2R^2}\right).$$

Because $C_{\text{Lev}} < 1$, this implies

$$m \leq \frac{R}{\sqrt{d-1}} \sqrt{2 \log \frac{1}{\alpha}} = r(\alpha).$$

Now

$$\mu(A_\varepsilon^c) = \mu(h > \varepsilon) \leq \mu(h > m + (\varepsilon - r(\alpha))_+),$$

and Lemma 1 yields exactly (2.4). $\square$

2.2 Balance implies nearby adversarial points

Definition 3 (Balanced classifier). *Let $F : \mathbb{S}^{d-1}(R) \rightarrow \{\pm 1\}$ be measurable, and write*

$$A_+ := \{x : F(x) = +1\}, \quad A_- := \{x : F(x) = -1\}.$$

For $p \in (0, 1/2]$, we say that F is p -balanced if

$$\mu(A_+) \geq p \quad \text{and} \quad \mu(A_-) \geq p.$$

Theorem 4 (Most points are close to both labels). *If F is p -balanced, then for every $\varepsilon > 0$,*

$$\mu((A_+)_\varepsilon \cap (A_-)_\varepsilon) \geq 1 - 2C_{\text{Lev}} \exp\left(-\frac{d-1}{2R^2} (\varepsilon - r(p))^2_+\right), \quad (2.5)$$

where $r(p)$ is defined in (2.3).

Proof. Apply Lemma 2 to A_+ and A_- separately and combine the two bounds by a union bound. $\square$

Theorem 5 (Adversarial neighbors for almost all points). *Let $F : \mathbb{S}^{d-1}(R) \rightarrow \{\pm 1\}$ be p -balanced with $p \in (0, 1/2]$. Fix $\delta \in (0, 1)$ and define*

$$\varepsilon_{\text{adv}}(p, \delta, d, R) := r(p) + \frac{R}{\sqrt{d-1}} \sqrt{2 \log \frac{2C_{\text{Lev}}}{\delta}}. \quad (2.6)$$

Then

$$\mu\left(\left\{x : \exists y \in \mathbb{S}^{d-1}(R) \text{ with } \|x - y\| \leq \varepsilon_{\text{adv}} \text{ and } F(y) \neq F(x)\right\}\right) \geq 1 - \delta.$$

In particular,

$$\varepsilon_{\text{adv}}(p, \delta, d, R) = O\left(\frac{R}{\sqrt{d}}(\sqrt{\log(1/p)} + \sqrt{\log(1/\delta)})\right).$$

Proof. Choose $\varepsilon = \varepsilon_{\text{adv}}$ in Theorem 4; then the right-hand side of (2.5) is at least $1 - \delta$. For every x in the intersection $(A_+)_{\varepsilon} \cap (A_-)_{\varepsilon}$, there are points $y_+ \in A_+$ and $y_- \in A_-$ with $\|x - y_{\pm}\| \leq \varepsilon$. One of them has label different from $F(x)$. $\square$

Remark 6. *The theorem separates two scales. The term $r(p) \asymp (R/\sqrt{d})\sqrt{\log(1/p)}$ is the price of imbalance; the term $(R/\sqrt{d})\sqrt{\log(1/\delta)}$ is the price of asking for success probability $1 - \delta$. In the exactly balanced case $p = 1/2$, only the second term remains.*

3 Oddness, threshold shifts, and level-set splitting

This section separates three distinct mechanisms for obtaining balance from a real-valued score f . The distinction matters, especially for ReLU networks.

3.1 Odd scores

Proposition 7 (Odd scores are balanced up to the zero level set). *Let $f : \mathbb{R}^d \rightarrow \mathbb{R}$ be odd, and restrict it to $\mathbb{S}^{d-1}(R)$. Then*

$$\mu(f > 0) = \mu(f < 0) = \frac{1}{2}(1 - \mu(f = 0)).$$

Consequently, any classifier obtained from f by thresholding at 0 is p -balanced with

$$p = \frac{1}{2}(1 - \mu(f = 0)).$$

If $\mu(f = 0) = 0$, then the classifier is exactly balanced.

Proof. The antipodal map $x \mapsto -x$ preserves μ and swaps $\{f > 0\}$ with $\{f < 0\}$. $\square$

3.2 When does a scalar threshold shift give exact balance?

Given a score $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ and a threshold $\theta \in \mathbb{R}$, define the two deterministic threshold classifiers

$$F_{\theta}^{-}(x) := \begin{cases} +1, & f(x) > \theta, \\ -1, & f(x) \leq \theta, \end{cases} \quad F_{\theta}^{+}(x) := \begin{cases} +1, & f(x) \geq \theta, \\ -1, & f(x) < \theta. \end{cases}$$

These are exactly the classifiers one obtains from a scalar bias shift together with one of the two uniform tie conventions on the zero set.

Theorem 8 (Exact balance by a genuine scalar threshold shift). *Let $\nu := f_{\#}\mu$ be the pushforward of μ under f . The following are equivalent.*

(i) *There exists $\theta \in \mathbb{R}$ such that F_{θ}^{-} or F_{θ}^{+} is exactly balanced.*

(ii) *There exists $\theta \in \mathbb{R}$ such that*

$$\nu((\theta, \infty)) = \frac{1}{2} \quad \text{or} \quad \nu([\theta, \infty)) = \frac{1}{2}.$$

(iii) *There exists $\theta \in \mathbb{R}$ such that*

$$\nu((-\infty, \theta)) = \frac{1}{2} \quad \text{or} \quad \nu((-\infty, \theta]) = \frac{1}{2}.$$

In particular, if ν is atomless, then there exists a threshold θ such that both F_{θ}^{-} and F_{θ}^{+} are exactly balanced.

Proof. By definition,

$$\mu(F_{\theta}^{-} = +1) = \mu(f > \theta) = \nu((\theta, \infty)), \quad \mu(F_{\theta}^{+} = +1) = \mu(f \geq \theta) = \nu([\theta, \infty)).$$

So (i) and (ii) are equivalent. Taking complements gives the equivalence with (iii). If ν is atomless, its distribution function is continuous and ranges from 0 to 1, so there exists θ with $\nu((-\infty, \theta]) = 1/2$. Atomlessness then gives $\nu(\{\theta\}) = 0$, hence both tails have mass 1/2. $\square$

Proposition 9 (Exact balance by measurable splitting of a median level set). *Let $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ be measurable and let θ be a median of f under μ . Then there exists a measurable set $B \subseteq \{f = \theta\}$ such that the classifier*

$$G_{\theta, B}(x) := \begin{cases} +1, & f(x) > \theta, \\ +1, & f(x) = \theta \text{ and } x \in B, \\ -1, & f(x) < \theta, \\ -1, & f(x) = \theta \text{ and } x \notin B \end{cases}$$

is exactly balanced.

Proof. Because θ is a median,

$$\mu(f > \theta) \leq \frac{1}{2}, \quad \mu(f < \theta) \leq \frac{1}{2}.$$

Hence

$$0 \leq \frac{1}{2} - \mu(f > \theta) \leq \mu(f = \theta).$$

Since μ is nonatomic on the sphere, there exists a measurable $B \subseteq \{f = \theta\}$ with

$$\mu(B) = \frac{1}{2} - \mu(f > \theta).$$

Then $\mu(G_{\theta, B} = +1) = 1/2$, and likewise $\mu(G_{\theta, B} = -1) = 1/2$. $\square$

Remark 10 (A sharp obstruction for ReLU-type scores). *For $0 < a < 1$, consider on $\mathbb{S}^{d-1}(1)$ the score*

$$f_a(x) := \text{ReLU}(x_1 - a) - \text{ReLU}(-x_1 - a).$$

Then $f_a > 0$ on a right cap, $f_a < 0$ on a left cap of the same measure, and $f_a = 0$ on the band $\{|x_1| \leq a\}$, which has positive measure. Each cap has measure strictly smaller than 1/2. Therefore no deterministic scalar threshold classifier F_{θ}^{-} or F_{θ}^{+} can be exactly balanced, although Proposition 9 still gives an exactly balanced measurable split of the level set $\{f_a = 0\}$. This is why level-set splitting must not be identified with an actual last-layer bias shift.

4 Applications to deep networks

4.1 Odd architectures

We first isolate the structural reason that many random architectures are exactly balanced.

Lemma 11 (Bias-free attention is odd). *Let $Q(x) = xW_Q$, $K(x) = xW_K$, and $V(x) = xW_V$ be linear maps with zero biases. Then*

$$\text{Attn}(x) := \text{softmax}(Q(x)K(x)^\top / \sqrt{d_k})V(x)$$

satisfies $\text{Attn}(-x) = -\text{Attn}(x)$.

Proof. We have $Q(-x) = -Q(x)$ and $K(-x) = -K(x)$, so the score matrix $Q(x)K(x)^\top$ is unchanged by $x \mapsto -x$. Hence the softmax weights are unchanged, while $V(-x) = -V(x)$. $\square$

Proposition 12 (Oddness from odd building blocks). *Any network obtained by composing the following operations is odd:*

1. *linear maps with zero biases;*
2. *coordinatewise odd activations;*
3. *LayerNorm with zero additive shift $\beta = 0$;*
4. *the attention block of Lemma 11;*
5. *residual additions $x \mapsto x + G(x)$ where G is odd.*

In particular, bias-free fully connected networks, bias-free CNNs, bias-free residual networks with odd activations, and bias-free ViT blocks built from these ingredients all compute odd logits.

Proof. Each listed operation is odd, and compositions and residual additions preserve oddness. $\square$

For random odd networks, exact balance follows once the zero set on the sphere is null. For analytic activations this is generic.

Proposition 13 (Generic null zero sets for analytic odd networks). *Fix $R > 0$. Suppose $\{f_w\}$ is a family of real-valued network outputs such that:*

1. *for every parameter vector w , the map $x \mapsto f_w(x)$ is real-analytic on $\mathbb{R}^d$;*
2. *the parameter law is absolutely continuous with respect to Lebesgue measure;*
3. *there exist $x_0 \in \mathbb{S}^{d-1}(R)$ and one parameter choice w_0 with $f_{w_0}(x_0) \neq 0$.*

Then, with probability one over w , the restriction of f_w to $\mathbb{S}^{d-1}(R)$ is not identically zero and

$$\mu(\{x \in \mathbb{S}^{d-1}(R) : f_w(x) = 0\}) = 0.$$

Proof. The map $w \mapsto f_w(x_0)$ is real-analytic and not identically zero, so its zero set has Lebesgue measure zero in parameter space. Hence with probability one, $f_w(x_0) \neq 0$, and therefore the restriction of f_w to the sphere is a nontrivial real-analytic function. A standard fact from real-analytic geometry says that the zero set of a nontrivial real-analytic function on a real-analytic manifold has measure zero. $\square$

Theorem 14 (Odd architectures are exactly balanced on the sphere). *Let f be the logit of any odd architecture covered by Proposition 12, and define $F(x) = \text{sign}(f(x))$ on $\mathbb{S}^{d-1}(R)$. If $\mu(f = 0) = 0$, then F is exactly balanced. Consequently, for every $\delta \in (0, 1)$,*

$$\mu\left(\left\{x : \exists y \in \mathbb{S}^{d-1}(R) \text{ with } \|x - y\| \leq \varepsilon_{\text{adv}}(1/2, \delta, d, R) \text{ and } F(y) \neq F(x)\right\}\right) \geq 1 - \delta.$$

In particular, the same conclusion holds almost surely for random analytic odd networks satisfying the hypotheses of Proposition 13.

Proof. Combine Proposition 7 with Theorem 5. □

Remark 15. *Theorem 14 should be read as a structural corollary of symmetry plus concentration, not as a new optimal-scaling theorem. For random CNNs, sharper constructive results are now available in Daniely [3], Daniely and Mehalel [4].*

4.2 Arbitrary scores, trained networks, and ReLU calibration

The geometric theorem applies to any classifier once balance is known. For a deployed network, the first question is therefore simply whether both labels occur on positive measure. The second question is whether exact balance can be achieved by a *genuine scalar threshold shift* of the logit. Theorem 8 gives the sharp answer.

Theorem 16 (What can be said for an arbitrary network logit?). *Let $f : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}$ be the logit of an arbitrary network, and let $\nu = f_{\#}\mu$. Then:*

(i) *If a deployed classifier $F : \mathbb{S}^{d-1}(R) \rightarrow \{\pm 1\}$ obtained from f already satisfies*

$$\mu(F = +1) > 0 \quad \text{and} \quad \mu(F = -1) > 0,$$

then F is p -balanced for

$$p := \min\{\mu(F = +1), \mu(F = -1)\} > 0,$$

and Theorem 5 gives adversarial neighbors for a $1 - \delta$ fraction of the sphere.

(ii) *If ν is atomless, then there exists a scalar threshold θ such that the genuine threshold classifier F_{θ}^{-} (equivalently F_{θ}^{+}) is exactly balanced, and Theorem 5 applies with $p = 1/2$.*

(iii) *Without any atomlessness assumption, every median θ admits an exactly balanced measurable split $G_{\theta,B}$ as in Proposition 9. This yields an exact balance statement for the score landscape, but it need not be realizable by a scalar bias shift of the deployed network.*

Proof. Part (i) is immediate from Theorem 5. Part (ii) is Theorem 8. Part (iii) is Proposition 9. □

Remark 17. *Theorem 16 is the corrected version of the earlier “median calibration” claim. Exact 50-50 balance produced by a true scalar threshold shift is available under the atomlessness condition in part (ii), but not in general; ReLU flat regions can obstruct it, as in Remark 10.*

4.3 Random Gaussian readouts

Let $a : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}^m$ be a measurable feature map and let

$$w \sim N(0, I_m/m), \quad f_w(x) := w^\top a(x).$$

Write

$$g(w) := \mu\{x : f_w(x) > 0\}.$$

We first record the exact first and second moments of $g(w)$.

Lemma 18 (Random Gaussian heads: expectation and variance). *Assume*

$$\mu(\{x : \|a(x)\| > 0\}) = 1. \tag{4.1}$$

For $X, X' \sim \mu$ independent, define

$$\rho(X, X') := \frac{\langle a(X), a(X') \rangle}{\|a(X)\| \|a(X')\|} \in [-1, 1] \quad a.s.$$

and

$$\Gamma(a) := \mathbb{E} |\rho(X, X')|.$$

Then

$$\mathbb{E}_w g(w) = \frac{1}{2}, \tag{4.2}$$

$$\text{Var}(g(w)) = \frac{1}{2\pi} \mathbb{E} [\arcsin \rho(X, X')], \tag{4.3}$$

$$\text{Var}(g(w)) \leq \frac{1}{4} \Gamma(a). \tag{4.4}$$

Proof. For fixed x with $a(x) \neq 0$, the scalar $w^\top a(x)$ is a nondegenerate centered Gaussian, so

$$\mathbb{P}_w(f_w(x) > 0) = \frac{1}{2}.$$

Integrating over x gives (4.2).

For the variance, write

$$g(w) = \int \mathbb{1}_{\{f_w(x) > 0\}} d\mu(x).$$

Then

$$\mathbb{E} g(w)^2 = \iint \mathbb{P}_w(f_w(x) > 0, f_w(x') > 0) d\mu(x) d\mu(x').$$

For fixed x, x' with nonzero features, the pair $(f_w(x), f_w(x'))$ is a centered bivariate Gaussian with correlation $\rho(x, x')$. Hence

$$\mathbb{P}_w(f_w(x) > 0, f_w(x') > 0) = \frac{1}{4} + \frac{1}{2\pi} \arcsin \rho(x, x').$$

Subtracting $(\mathbb{E} g(w))^2 = 1/4$ yields (4.3). Finally, $|\arcsin r| \leq (\pi/2)|r|$ on $[-1, 1]$, giving (4.4). $\square$

A second issue is nontriviality: we need both labels to occur on positive measure. The following deterministic condition is stronger and cleaner than the earlier draft's false measure-theoretic cone condition.

Proposition 19 (A geometric nontriviality criterion for readouts). *Assume that a is continuous and*

$$0 \in \text{int conv } a(\mathbb{S}^{d-1}(R)). \quad (4.5)$$

Then for every nonzero $w \in \mathbb{R}^m$,

$$\mu\{x : w^\top a(x) > 0\} > 0 \quad \text{and} \quad \mu\{x : w^\top a(x) < 0\} > 0.$$

Proof. Fix $w \neq 0$. If $w^\top a(x) \geq 0$ for every x , then $a(\mathbb{S}^{d-1}(R))$ and therefore its convex hull lie in the closed half-space $\{v : w^\top v \geq 0\}$. Then 0 cannot be an interior point of the convex hull, contradicting (4.5). So there exist x_+, x_- with $w^\top a(x_+) > 0$ and $w^\top a(x_-) < 0$. By continuity, the same inequalities hold on neighborhoods of x_+ and x_- , and every nonempty open set on the sphere has positive μ -measure. $\square$

We also need the zero level set to be null so that ties do not matter.

Lemma 20 (Random Gaussian readouts have null zero sets). *Under (4.1),*

$$\mathbb{P}_w(\mu(\{x : f_w(x) = 0\}) = 0) = 1.$$

Proof. For each fixed x with $a(x) \neq 0$, the random variable $f_w(x)$ is nondegenerate Gaussian, so $\mathbb{P}_w(f_w(x) = 0) = 0$. Fubini's theorem gives

$$\mathbb{E}_w \mu\{x : f_w(x) = 0\} = \int \mathbb{P}_w(f_w(x) = 0) d\mu(x) = 0.$$

The claim follows because the integrand is nonnegative. $\square$

Corollary 21 (Near-balance for random Gaussian heads). *Assume (4.1) and the hypotheses of Proposition 19. Then for every $\tau > 0$,*

$$\mathbb{P}_w(|g(w) - \tfrac{1}{2}| \geq \tau) \leq \frac{\Gamma(a)}{4\tau^2}.$$

Moreover, with probability one, $\mu(f_w = 0) = 0$ and both labels have positive measure. Hence on the event $|g(w) - 1/2| < \tau$, the threshold classifier $x \mapsto \text{sign}(f_w(x))$ is p -balanced with

$$p \geq \frac{1}{2} - \tau.$$

Proof. The probability bound is Chebyshev together with Lemma 18. The positivity of both class masses follows from Proposition 19 since $w \neq 0$ almost surely. Lemma 20 removes ties. $\square$

Theorem 22 (Adversarial neighbors for random Gaussian heads). *Under the assumptions of Corollary 21, fix $\tau \in (0, 1/2)$ and $\delta \in (0, 1)$. Then with probability at least $1 - \Gamma(a)/(4\tau^2)$ over the draw of w , the classifier $x \mapsto \text{sign}(f_w(x))$ satisfies the conclusion of Theorem 5 with $p \geq 1/2 - \tau$. In particular, on that event a $1 - \delta$ fraction of inputs admit an adversarial neighbor within radius*

$$\varepsilon_{\text{adv}}(1/2 - \tau, \delta, d, R).$$

Proof. Combine Corollary 21 with Theorem 5. $\square$

Corollary 23 (High-dimensional regime). *For each dimension d , let $a_d : \mathbb{S}^{d-1}(R) \rightarrow \mathbb{R}^{m_d}$ satisfy the assumptions of Theorem 22, and define $\Gamma_d := \Gamma(a_d)$. If $\Gamma_d \log d \rightarrow 0$, then for every fixed $\gamma > 0$ there exists a sequence of radii*

$$\varepsilon_d = O\left(\frac{R\sqrt{\log d}}{\sqrt{d}}\right)$$

such that, when $w_d \sim N(0, I_{m_d}/m_d)$ and $X_d \sim \mu_d$ are independent,

$$\mathbb{P}(X_d \text{ has no } \varepsilon_d\text{-adversarial neighbor for } \text{sign}(w_d^\top a_d)) \rightarrow 0.$$

Proof. Take $\tau_d := 1/\sqrt{\log d}$ and $\delta_d := d^{-\gamma}$. Corollary 21 shows that the near-balance event fails with probability at most $\Gamma_d \log d/4$. On the near-balance event, Theorem 5 with $p \geq 1/2 - \tau_d$ and failure parameter δ_d gives adversarial neighbors for all but a δ_d fraction of the sphere. Since $r(1/2 - \tau_d) = O(R/\sqrt{d})$, the resulting radius is $O(R\sqrt{\log d}/\sqrt{d})$. The total failure probability is at most $\Gamma_d \log d/4 + \delta_d \rightarrow 0$. $\square$

5 Transfer to non-uniform measures on the same sphere

The geometric theorem is stated for the uniform spherical measure μ . A simple domination argument transfers it to any other distribution with bounded density with respect to μ .

Theorem 24 (Measure transfer). *Let ν be a probability measure on $\mathbb{S}^{d-1}(R)$ such that $\nu \ll \mu$ and*

$$\frac{d\nu}{d\mu} \leq \rho_{\max} \quad \mu\text{-a.e.}$$

for some finite $\rho_{\max}$. Suppose a classifier F satisfies the conclusion of Theorem 5 for some radius ε and failure parameter δ , i.e.

$$\mu(E_\varepsilon(F)) \geq 1 - \delta,$$

where

$$E_\varepsilon(F) := \left\{x : \exists y \in \mathbb{S}^{d-1}(R) \text{ with } \|x - y\| \leq \varepsilon \text{ and } F(y) \neq F(x)\right\}.$$

Then

$$\nu(E_\varepsilon(F)) \geq 1 - \rho_{\max}\delta.$$

Proof. Since $\nu(E_\varepsilon(F)^c) = \int_{E_\varepsilon(F)^c} (d\nu/d\mu) d\mu$, the density bound gives

$$\nu(E_\varepsilon(F)^c) \leq \rho_{\max}\mu(E_\varepsilon(F)^c) \leq \rho_{\max}\delta.$$

$\square$

Corollary 25 (Non-uniform spherical data). *If F is p -balanced with respect to the uniform spherical measure μ , then for every $\delta \in (0, 1)$ and every $\nu \ll \mu$ with density bounded by $\rho_{\max}$,*

$$\nu(E_{\varepsilon_{\text{adv}}(p, \delta, d, R)}(F)) \geq 1 - \rho_{\max}\delta.$$

Proof. Combine Theorems 5 and 24. $\square$

Remark 26. *The same two-step template applies on other reference spaces once a concentration inequality is available there. In particular, one can formulate ball-level analogues using known concentration results for the uniform measure on $B(0, R)$, but we do not pursue that extension here. This keeps the paper focused on the spherical argument, which is the cleanest setting for the balance viewpoint.*

6 Discussion

The message of this note is intentionally modest. The optimal adversarial scale $R/\sqrt{d}$ was already known in several random-network settings and from earlier geometric work. What this paper adds is a cleaner bookkeeping device: the balance parameter p . Once p is understood, spherical concentration does the rest.

There are three points worth highlighting. First, odd symmetry is a powerful and very simple source of exact balance. Second, for general logits one must distinguish genuine scalar threshold shifts from abstract measurable splits of a level set; this distinction is essential for ReLU networks. Third, random Gaussian readouts are naturally centered at balance, and the deviation from exact balance is controlled by a single feature-correlation parameter.

Several directions remain open. A more refined theory for ReLU networks would quantify when the pushforward score distribution is atomless, or at least when median atoms are small. On the random-head side, it would be interesting to replace the Chebyshev bound by genuinely sub-Gaussian concentration under natural regularity assumptions on the feature map. Finally, extending the same balance-based packaging to other geometries and data models, while keeping the claims honest about what is geometric and what is architecture-specific, remains worthwhile.

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9 Declaration of generative AI and AI-assisted technologies in the manuscript preparation process.

During the preparation of this work the author(s) used ChatGPT pro, in order to speed up the process of writing. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

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